

Physics 215 Course Policies and Syllabus

U.S. Air Force Academy, Department of Physics and Meteorology

I. COURSE DESCRIPTION

Physics 215. *General Physics II with Laboratory*. Introductory calculus-based physics course with emphasis on contemporary applications (Part II). Topics include electrostatics, simple DC circuits, magnetic fields, electromagnetic induction, electromagnetic waves, and physical optics. Possible additional topics include simple AC circuits and applications, geometric optics, and selected topics in modern physics. Course emphasizes the use of vectors and calculus in problem solving. Course includes in-class laboratories and computer applications to highlight key concepts. Highly desirable for course to be taken in the semester immediately following the successful completion of Physics 110.

II. COURSE PREREQUISITES

You must have completed (or validated) Physics 110 and Math 142/152. You are expected to be knowledgeable and proficient in the following math concepts/skills:

- algebra and trigonometry
- vector operations, including dot product and cross product
- differentiation of polynomials and simple functions
- integration of polynomials and simple functions

III. COURSE GOALS

Core physics courses, including Physics 110 and Physics 215, contribute to the development and assessment of the *Scientific Reasoning and the Principles of Science* outcome, to apply scientific habits of mind, including proficiency in the nature of science, scientific reasoning, and the principles of science. This includes the following general proficiencies:

1. *Nature of Science* – Determine if explanations, claims, or statements are consistent with the Nature of Science and explain how a scientific explanation can simultaneously be broadly accepted as reliable yet provisional.
2. *Scientific Reasoning* – Use if-then reasoning to generate a scientifically testable expectation; describe the effects of data collection methods on the reliability of results; use mathematics and appropriate technology to describe and represent data (e.g., symbolically, visually, or numerically); and systematically interpret results to draw valid conclusions.
3. *Principles of Science* – Identify, explain, and discuss the role of foundational science principles in real-world applications and apply foundational principles of science and mathematics to predict or explain natural phenomenon

Additionally, during Physics 215 you will learn to:

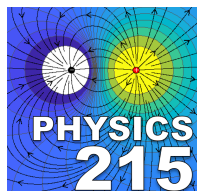
1. Identify how the fundamental physical principles of electricity and magnetism apply to conceptual or quantitative problems.
2. Solve conceptual or quantitative physical problems involving electricity, magnetism, and modern physics.
3. Apply experimental skills to investigate the physical principles governing electricity and magnetism.
4. Analyze and explain the physical principles that apply to the operation of electromagnetic systems and circuits.

IV. COURSE ADMINISTRATION

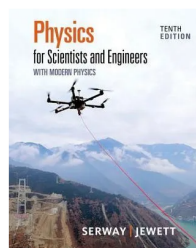
Head, Department of Physics and Meteorology: Col Hsien-Liang “Rose” Tseng
 Director of Core Physics: LTC Tyler Jones
 Physics 215 Course Director: Lt Col Benjamin Roth
 Physics 215 Assistant Course Director: Lt Col Casey Pellizzari

V. COURSE MATERIAL

Each cadet is required to purchase website access to Cengage WebAssign, which comes with access to an eBook version of the textbook. Students are encouraged, but not required, to purchase a hard copy of the text. Your Cengage WebAssign account will be linked to this course by logging into our Course BlackBoard and clicking on assignments. Course announcements will be sent through the “Physics 215 Fall 2026” team in Microsoft Teams. Any other required materials (i.e., lab handouts, supplemental worksheets) will be available in the ‘Files’ folder of the Teams channel.



Physics 215 Team



Physics for Scientists and Engineers

- **WebAssign** (REQUIRED) eBook, assignments (preflights and homework), and additional resources (worked examples, instructional videos, etc.) through Cengage.
 - WebAssign ISBN-13: 9781337699280
- **Textbook** (ENCOURAGED) R. A. Serway and J. W. Jewett, Jr., *Physics for scientists and engineers* (Cengage Learning, 2018).
 - Hardcover ISBN-13: 9781337553292
 - Loose-leaf Edition ISBN-13: 9781337553452

VI. GRADED EVENTS

Graded Event	Number	Points (Each)	Total (%)
Preflight	37	2	74 (7.2%)
Homework	31	2	62 (6.1%)
Labs	6	20	120 (11.7%)
Exam-Prep Quiz	7	20	140 (13.7%)
Cumulative Homework Set	4	7	28 (2.7%)
Graded Reviews	3	100	300 (29.3%)
Final Exam	1	300	300 (29.3%)
Total			1024 (100%)

NOTE: A sufficiently low grade (less than 40%) on the final exam could result in failure of the course regardless of the overall score. At the discretion of the Physics and Meteorology Department Head, any final exam score below 40% will result in an evaluation committee review to determine if you will pass, regardless of your overall course average.

VII. COURSE WORK

Most lessons in this course include two out-of-class assignments: pre-class preparation in the form of a short reading and preflight assignment, and post-class homework. You should expect to spend approximately an hour on each of these components. It is important that you put meaningful effort into these tasks. If you are spending significantly more time than this for each lesson you should seek EI from your instructor. Successful completion of the preflight and homework assignments is vital for your learning and accounts for approximately 20% of your overall grade. If you earn an average of 80% on your labs, EPQs, and GRs, but only complete 50% of your preflights and homework, you will go into the final exam with a 72% in the course. If you instead earn 95% on

your preflights and homework, you will go into the final exam with an 84% in the course. Giving up these points by missing assignments will have a direct negative impact on your grade, amplified because less effort spent working through these assignments will likely lead to lower EPQ, GR and Final scores.

Reading

There will be a reading assignment to complete every lesson as noted on the syllabus.

Preflight

The Pre-lesson Readiness Engagement Platform (PREP) and Interactive Pre-lesson Readiness Engagement Platform (iPREP) will serve as the preflight platforms for this course. PREP allows you to answer short conceptual questions about the assigned pre-class materials, while iPREP engages you in an interactive dialogue about the lesson content. These assignments are designed to introduce key concepts, assess your understanding of the assigned material, and provide feedback to your instructor before class. You will receive credit based on participation, effort, scientific reasoning, and critical thinking rather than solely on the correctness of your answers. Links to PREP and iPREP will be available through the course Blackboard site. Both platforms are entirely web-based and do not require you to install any software. Using iPREP requires either a free or paid Claude account. Unless otherwise specified, preflight assignments are due by 2359 on the night before the lesson is taught to your section. Each preflight assignment is worth 2 points.

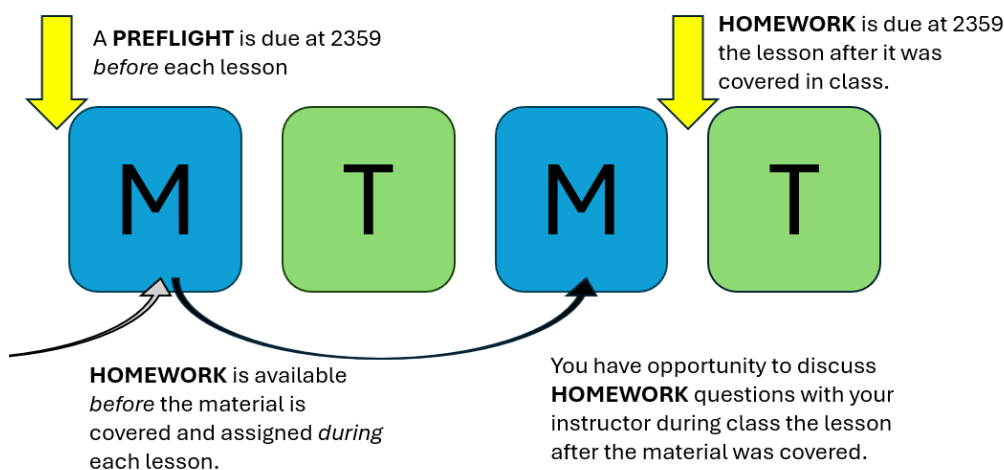


FIG. 1. Preflight and homework battle rhythm.

Math Skills Enhancement (MSE)

During the semester five MSE exercises will be offered. MSE are an opportunity to practice math skills that are critical to Physics 215 material. Each MSE will be scheduled to augment class material at that point of the course, and will help cement important math skills without additional Physics content. Each MSE is worth 2 extra credit points. MSE exercises are scheduled and available in WebAssign. MSE Topics include algebra, trigonometry, vectors, dot products, cross products, and differentiation. Each set is mastery based, and will require 4 of out 6 questions to be answered correctly. MSE exercises are due Sunday night at 2359:

MSE Exercise	Due Date (2359)
1	9 Aug
2	23 Aug
3	30 Aug
4	10 Sept
5	11 Oct

Homework

With the exceptions annotated on the schedule, every lesson will have a homework assignment. Homework is your third exposure to the lesson material and is meant to solidify conceptual understanding and problem-solving skills. Homework for a specific lesson will become available prior to that lesson and must be completed via WebAssign before 2359 the lesson after it was covered in class or on the following Sunday if the lesson after was a Friday (see FIG. 1). This allows time for you to ask your instructor clarifying homework questions the following lesson. Each homework set is worth 2 points. Each question will have 10 attempts, with 25% of points deducted for each attempt after 5.

Additionally, before each graded review and the final there is a cumulative homework set, indicated by a **C** on the syllabus. This problem set serves as a refresher for material presented throughout the block, but will not cover all testable material (i.e., this should *not* be your only GR or final study). Each review set is worth 7 points and is due at 2359 before each GR.

Labs

Each lab will include a WebAssign preflight assignment to be completed by 2359 the night before you conduct the lab. On lab days, classes will span the full double period. During class on lab days, your lab group will apply experimental skills to investigate electromagnetic principles and complete data collection and analysis. Each lab is worth 20 points. Labs are due 2359 the following non-GR lesson after completing the lab in class. Lab reports will be progressive with the goal of developing skills to effectively communicate lab work into a well organized scientific report.

Exam-Prep Quizzes

Exam-Prep Quizzes (EPQs) consist of multiple-choice and workout problems like those on GRs and the final exam. They will be administered during class time, and you will have 20 minutes to complete each EPQ. You should use these quizzes to gauge your understanding of the material before the exams. There will be 7 EPQs, each worth 20 points.

Graded Reviews

Graded Reviews (GRs) typically consist of multiple-choice questions, short answer questions, and workout problems. They will be administered during the common GR period, and you will have 50 minutes to complete each GR. There will be 3 GRs, each worth 100 points.

Final Exam

There will be a comprehensive final exam at the end of the semester. The final exam is your opportunity to demonstrate your mastery of the material. As such, validation of the final exam is not offered. The final exam is worth 300 points.

VIII. DOCUMENTATION POLICY AND UNAUTHORIZED RESOURCES

DFPM's documentation standards align with DF policy. DFPM encourages collaboration and group work to help you master challenging concepts. Department policies regarding unauthorized resources are focused on eliminating shortcuts and workarounds that have been shown to hinder the learning process. In general, **unauthorized resources** are resources that do not help you learn, but rather provide an answer to a problem without developing the understanding of how to solve it. **Unauthorized resources should not be used when completing assignments for a grade.**

- **Authorized resource:** Asking your roommate for assistance solving a homework problem because you have attempted it and were confused on a step.
- **Unauthorized resource:** Asking your roommate for the numerical solution to a homework problem.

Generative AI or solution providing website usage guidance. Based on the USAFA GenAI guidance memo, this course allows level 3 usage of GenAI, including use on lab assignments. AI platforms and solution providing websites can be either authorized or unauthorized resources, depending on how you interact with them. Using GenAI to guide you through the physics of a problem, much like you would use a textbook, is authorized. Among the benefits of AI is the ability to generate an explanation of a concept and then ask follow-on questions to define terms or expand on parts of the explanation. Using these tools in this fashion will generally enhance learning and DFPM encourages exploration of this technology. Looking at a generated solution or directly asking an AI platform for a solution to a problem on a graded assignment (just like asking someone to do your homework for you) is unauthorized. Technology used in this way is not effective for learning and can too easily circumvent the learning that takes place when you actually work through an assigned problem. **Solution-providing websites such as Chegg.com are not authorized resources for core physics courses.**

EPQs and GRs are **individual effort**; all other assignments are **collaborative**. You do not need to document use of your course textbook, class slides/handouts, or assistance you receive from your instructor, but always document all other resources you reference as you complete graded assignments. Regardless of what resources you use, or how you use them, accurately describe the level of assistance received. Assignments completed using an unauthorized resource will not receive academic credit, but proper documentation is essential to avoid any concerns about honor or academic integrity. If you have any questions about authorized or unauthorized resources, please discuss with your course instructor.

IX. ABSENCES, TARDINESS, AND MAKE-UP EXAMS

If you will miss a lesson for any reason, you will still be expected to complete all preflight and homework assignments on the due date, unless you have been granted an extension by your instructor.

Squadron Commander Bedrest

Students on Squadron Commander bedrest must e-mail their instructor AND copy their Squadron Commander and AMT *before* missing class to show that their bedrest is approved. Otherwise, the student will be marked absent and any missed work will be subject to a 25% penalty.

Labs

If you will be absent or more than 20 minutes late, you will be required to turn in your own lab worksheet or report. For excused absences, you must complete missed labs within 3 lessons (6 business days) of the originally scheduled date. You alone are responsible for scheduling a time to make up any missed in-class activity, regardless of the reason. Exceptions to this policy must be approved by the course director. *Labs are subject to a 2 point deduction from their total score for each day they are late.*

Graded Reviews

If you will be absent during a GR due to a Scheduling Committee Action (SCA), you are responsible to notify your instructor at least THREE BUSINESS DAYS (i.e., not including weekends) PRIOR to the offering of the exam. If you are more than 15 minutes late unexcused for a GR, you must take a makeup exam with a 25% penalty. If you are less than 15 minutes late, you may still take the exam during the remaining scheduled time.

Make-up Exams (GRs and EPQs)

Make-up exams will only be offered without penalty for students absent with an excused absence. Exams will not be administered on the road unless approved by the Course Director on a case-by-case basis. Work with your instructor to schedule a time to make up the exam or quiz within two lessons.

X. EXTRA INSTRUCTION (EI)

If you are having difficulty understanding course concepts during classroom contact hours, you are encouraged to seek EI with your instructor, other DFPM faculty members, or Quantitative Reasoning Center (QRC) professionals. Most lessons are 90 minutes in duration with the final 23 minutes of the double period reserved as your primary opportunity for EI. Your individual course instructor will provide their policy regarding EI.

XI. LEARNING OBJECTIVES**Lab Objectives**

- L-1** Develop lab and data collection techniques suitable for analysis and reporting results.
- L-2** Analyze the quantitative results of an experiment and develop appropriate conclusions about the physical world using mathematical techniques.
- L-3** Develop symbolic expressions that accurately depict a theoretical model being examined by an experiment.
- L-4** Understand the impact of measurement uncertainty and error in experimentation and develop experimental and statistical mitigation techniques.
- L-5** Develop confidence in the scientific method as an approach to investigate everyday phenomenon.
- L-6** Communicate scientific reasoning and experimental data through structured, clear technical writing and appropriate visual representations.

Scientific Method

- 1-1** Explain the scientific process of hypothesis development, testing, and peer-review.

1-2 Analyze observations and data to develop conclusions.

1-3 Apply the scientific method in a laboratory setting.

Electric Charge and Forces

1-4 Explain the concepts of displacement, direction, and distance between locations using vector and scalar quantities.

1-5 Describe the nature of electric charge.

1-6 Describe the effect charged objects have on neutral objects and other charged objects.

1-7 Calculate the magnitude and direction of electric forces between point charges using Coulomb's law.

Electric Fields

1-8 Explain how electric fields are formed and how they are related to electric forces.

1-9 Construct electric field lines and interpret electric field line diagrams.

1-10 Calculate the magnitude and direction of electric fields due to discrete and simple continuous charge distributions.

Matter in Electric Fields

1-11 Quantitatively predict the motion of charged particles inside electric fields using Newton's second law, kinematics, and/or work-energy principles.

Electric Flux and Gauss's Law

1-12 Explain the concept of electric flux and how it relates to electric field.

1-13 Use Gauss's law to calculate electric fields with spherical, cylindrical and planar symmetry.

Electric Potential

1-14 Explain the concept of electric potential and how it relates to electric field.

1-15 Explain the relationship between electric potential and potential energy.

1-16 Calculate the electric potential at a location in space due to discrete charge distributions.

1-17 Experimentally measure the electric potential.

Electrostatic (Potential) Energy

1-18 Calculate the electrostatic potential energy of discrete charge distributions.

1-19 Explain the relationship between electrostatic potential energy and work.

Capacitance

- 2-1 Explain the concept of capacitance of a capacitor and how it relates to the physical design of the capacitor.
- 2-2 Describe how capacitors store electrical energy.

Electric Current

- 2-3 Explain the concept of electric current, explain why charged particles move and how the physical properties of the conductor determine the amount of current flow.
- 2-4 Explain the concept of electrical resistance and describe the relationship between electric potential difference, electric current, and resistance using Ohm's law.
- 2-5 Interpret and calculate the electrical power dissipated in the circuits analyzed.
- 2-6 Analyze direct-current circuits with equivalent resistance and Kirchhoff's rules.
- 2-7 Experimentally measure the values of electric current, voltage and resistance in a circuit.
- 2-8 Analyze the charging and discharging of a capacitor in an RC circuit.

Magnetic Forces

- 2-9 Explain the concept of magnetic force and how it relates to magnetic field.
- 2-10 Calculate the magnitude and direction of the magnetic force on a moving charged particle in the presence of a magnetic field.
- 2-11 Calculate the magnitude and direction of the magnetic force on a current-carrying wire in the presence of a magnetic field.
- 2-12 Quantitatively predict the motion of charged particles inside magnetic fields using Newton's second law, kinematics, and/or work-energy principles.

Magnetic Fields

- 2-13 Explain the origin of magnetic fields and recognize that electric fields and magnetic fields can both concurrently affect the motion of charged particles.
- 2-14 Use the Biot-Savart law to calculate and sketch the magnetic fields created by various current geometries; recognize the limitations of the simplified versions of the Biot-Savart law.
- 2-15 Use Ampere's law to determine magnetic fields of symmetric current distributions and recognize its limitations.

Magnetic Dipoles

- 2-16 Explain the concepts of magnetic dipole and magnetic dipole moment.
- 2-17 Explain the concept of magnetic flux and understand the significance of Gauss's law for magnetism.
- 2-18 Calculate the magnitude and direction of the magnetic dipole moment and the torque felt by a current loop in an external magnetic field.

Electromagnetic Induction

- 3-1 Explain the phenomenon of electromagnetic induction, how magnetic flux can change, and how induction is consistent with conservation of energy.
- 3-2 Use Faraday's law to calculate induced emfs and electric currents; use Lenz's law to determine the directions of induced currents.
- 3-3 Explain quantitatively how an electric generator converts mechanical energy into electric energy.
- 3-4 Use Faraday's law to calculate an induced electric field.

Alternating Current

- 3-5 Describe alternating current and how a generator is used to create alternating current.
- 3-6 Calculate the input and output characteristics of transformers such as voltage, current and number of turns.

Maxwell's Equations

- 3-7 Explain Maxwell's equations as the four fundamental principles governing all electromagnetic phenomena.
- 3-8 Explain how Maxwell's inclusion of displacement current completed the set of unifying electromagnetic principles.
- 3-9 Explain how Maxwell's versions of Faraday's law and Ampère's law laid the foundation for the theory of electromagnetic waves.

Electromagnetic Waves and Light

- 3-10 Interpret the mathematical expressions describing wave motion (concepts such as amplitude, wave period, wavelength, wave number, and phase angle).
- 3-11 Describe the origin, shape, direction of propagation, and relationship of the electric and the magnetic fields of an electromagnetic wave propagating through space.
- 3-12 Identify the frequencies and wavelengths of various regions of the electromagnetic spectrum.
- 3-13 Explain the electromagnetic wave propagation between regions of varying index of refraction, including Snell's law and total internal reflection.

Images from Mirrors and Lenses

- 3-14 Explain how the particle model of light is used to create geometric images of observable objects.
- 3-15 Using the paraxial mirror and thin lens approximations, quantitatively describe the location, size, and orientation of an image produced from a simple optical system.
- 3-16 Apply geometric ray tracing techniques to simple lenses in order to find the image of the object.
- 3-17 Calculate and explain the process to find the image of an object through a combination of thin lenses.

Interference

- 4-1 Explain why coherent and monochromatic light is often used to study the phenomena of interference and diffraction of light.
- 4-2 Explain the concepts of constructive and destructive interference.
- 4-3 Quantitatively analyze interference patterns.

Diffraction

- 4-4 Quantitatively analyze single-slit diffraction patterns.
- 4-5 Explain how diffraction can limit our ability to form resolvable images with certain optical systems.
- 4-6 Use the Rayleigh criterion to analyze simple optical systems with specified resolution and wavelength/aperture characteristics.
- 4-7 Explain phenomena that result in the polarization of light waves and the analysis of these waves with Malus's Law and Brewster's angle.

Modern Physics

- 4-8 Explain the photoelectric effect in terms of the quantized nature of electromagnetic waves; perform calculations of photon energy, work function and electron kinetic energy.
- 4-9 Qualitatively describe the wave-particle duality of light and matter.
- 4-10 Describe the concepts of radioactivity, fission, and fusion.

Lsn	M Day	T Day	Topic	Reading	PF	In-Class	HW	
1	Thu 6 Aug	Fri 7 Aug	Course Admin/ Vector Review	3.1–3.4, 22.1	N	CSEM	N	Elec. Charge & Fields
2	Mon 10 Aug	Tue 11 Aug	Electric Charge, Coulombic Force	22.2–22.3	Y		Y	
3	Wed 12 Aug	Thu 13 Aug	Coulomb's Law, Lab Techniques	22.3, Lab Manual	Y		Y	
4	Fri 14 Aug	Mon 17 Aug	Electric Fields and Superposition	22.4–22.5	Y		Y	
5	Tue 18 Aug	Wed 19 Aug	Charged Particles in Uniform Elec. Fields	22.6	Y	EPQ1	Y	
6	Thu 20 Aug	Fri 21 Aug	LAB: Quantized Charge	Lab Handout	Y	LAB1	N	Elec. Potential
7	Mon 24 Aug	Tue 25 Aug	Charge Distributions, Electric Flux	23.1–23.2	Y		Y	
8	Wed 26 Aug	Thu 27 Aug	Gauss's Law and Its Applications	23.3–23.4	Y		Y	
9	Fri 28 Aug	Mon 31 Aug	Electric Potential Difference	24.1–24.2	Y		Y	
10	Tue 1 Sep	Wed 2 Sep	Electric Potential, Potential Energy	24.3–24.4, 24.6	Y	EPQ2	Y	
11	Thu 3 Sep	Fri 4 Sep	LAB: Mapping Electric Potential	Lab Handout	Y	LAB2	C	Circuits
12	8 September		GRADED REVIEW 1		N		N	
13	Thu 10 Sep	Mon 14 Sep	Capacitance, Energy, and Dielectrics	25.1–2, 25.4–5	Y		Y	
14	Tue 15 Sep	Wed 16 Sep	Current, Resistance, and Electrical Power	26.1–26.2, 26.6	Y		Y	
15	Thu 17 Sep	Mon 21 Sep	DC Circuit Analysis, Kirchhoff's Rules	27.1–27.3	Y		Y	
16	Tue 22 Sep	Wed 23 Sep	RC Circuits & Pre-Lab Activity	27.4	Y		Y	Mag. Fields
17	Thu 24 Sep	Fri 25 Sep	LAB: Building DC Circuits	Lab Handout	Y	LAB3	N	
18	Mon 28 Sep	Tue 29 Sep	Moving Charged Particle in a Magnetic Field	28.1–28.2	Y	EPQ3	Y	
19	Wed 30 Sep	Thu 1 Oct	Magnetic Force on Current-carrying Wires	28.3–28.4	Y		Y	
20	Fri 2 Oct	Mon 5 Oct	Magnetic Dipoles and Torque	28.5	Y		Y	
21	Tue 6 Oct	Wed 7 Oct	Sources of Magnetic Fields	29.1–29.2	Y	EPQ4	Y	Changing Fields
22	Thu 8 Oct	Fri 9 Oct	Ampère's Law, Gauss's Law in Magnetism	29.3–29.5	Y		C	
23	14 October		GRADED REVIEW 2		N		N	
24	Thu 15 Oct	Fri 16 Oct	Faraday's Law of Induction, Motional EMF	30.1–30.2	Y		Y	
25	Mon 19 Oct	Tue 20 Oct	Lenz's Law, Induced Electric Field	30.3–30.4	Y		Y	
26	Wed 21 Oct	Thu 22 Oct	Generators and Motors, AC, Transformers	30.5, 32.1–2, 32.8	Y		Y	Optics & Nature of Light
27	Fri 23 Oct	Mon 26 Oct	LAB: Building an Electric Motor	Lab Handout	Y	LAB4	N	
28	Tue 27 Oct	Wed 28 Oct	Displacement Current	33.1–33.2	Y	EPQ5	Y	
29	Thu 29 Oct	Fri 30 Oct	Maxwell's Equations	33.2	Y		Y	
30	Mon 2 Nov	Tue 3 Nov	Electromagnetic Waves, EM Spectrum	33.3, 33.4, 33.7	Y	CSEM	Y	
31	Wed 4 Nov	Thu 5 Nov	Light, Reflection, Refraction	34.1–34.4, 34.7	Y	EPQ6	Y	Modern
32	Fri 6 Nov	Mon 9 Nov	Image Formation from Mirrors	35.1–35.2	Y		Y	
33	Tue 10 Nov	Thu 12 Nov	Image Formation from Lenses	35.3–35.4	Y		Y	
34	Fri 13 Nov	Mon 16 Nov	LAB: Thin Lenses	Lab Handout	Y	LAB5	C	
35	18 November		GRADED REVIEW 3		N		N	
36	Thu 19 Nov	Fri 20 Nov	Double-Slit Interference	36.1–36.2	Y		Y	
37	Mon 23 Nov	Tue 24 Nov	Diffraction, Resolution	37.1–37.3	Y		Y	
38	Tue 1 Dec	Wed 2 Dec	LAB: Single/Double-Slit Diffraction	Lab Handout	Y	LAB6	N	
39	Thu 3 Dec	Fri 4 Dec	Intro to Nuclear (Planetarium)	43.8, 43.10	Y		Y	
40	Mon 7 Dec	Tue 8 Dec	Polarization	37.6	Y	EPQ7	Y	
41	Wed 9 Dec	Thu 10 Dec	Photoelectric Effect, Wave/Particle	34.1, 39.2, 39.4	Y		C	
FINAL EXAM Period TBD								